DELHI PUBLIC SCHOOL

Subject: Physics **Class: 9**

Time Allowed: 60 minutes Maximum Marks: 60

All questions are compulsory unless stated otherwise.

Name: ______________________ Roll Number: __________________

Section: ______________

GENERAL INSTRUCTIONS:

1. All questions are compulsory.
2. Write your answers in the space provided.
3. Read each question carefully before answering.
4. Marks are indicated against each question.

Attempt all questions. Write clearly and concisely.

SECTION A

Attempt 4 out of 4 questions

Conceptual Understanding

Attempt all questions

1. Define inertia and provide one common example from daily life illustrating its effect.(2 Marks)

2. State Newton's First Law of Motion in your own words, clearly mentioning its other name.(2 Marks)

3. Fill in the blank: The product of an object's mass and its velocity is a vector quantity known as __________, and its SI unit is __________.(2 Marks)

4. True or False: According to Newton's Third Law, action and reaction forces always act on the same body, leading to their cancellation. Justify your answer briefly.(2 Marks)

SECTION B

Attempt 7 out of 9 questions

Application Based Questions

Attempt any questions

1. Explain why passengers tend to fall forward when a moving bus suddenly applies brakes. Which law of motion is primarily demonstrated in this scenario?(3 Marks)

2. Differentiate between balanced and unbalanced forces, giving a practical example for each type of force to illustrate their effects.(3 Marks)

3. A bullet of mass 20 g is horizontally fired from a gun of mass 2 kg with a velocity of 150 m/s. Calculate the recoil velocity of the gun, assuming the gun was initially at rest.(3 Marks)

4. Describe how Newton's second law of motion provides a quantitative measure of force. Write down its mathematical formulation and explain the terms involved.(3 Marks)

5. Why is it significantly easier to stop a tennis ball than a cricket ball moving with the exact same speed? Explain your reasoning in terms of a relevant physical quantity.(3 Marks)

6. An astronaut pushing against a wall inside a spaceship experiences an equal and opposite force. Which law of motion governs this interaction? Provide another distinct practical example of this law.(3 Marks)

7. State the law of conservation of momentum. Provide a simple scenario where this law is clearly observed, such as during a collision.(3 Marks)

SECTION C

Attempt 4 out of 5 questions

Critical Thinking Questions

Attempt any questions

1. A car of mass 1500 kg is travelling at a speed of 90 km/h. When the brakes are applied, it comes to a complete stop in 4 seconds. Calculate the magnitude of the force exerted by the brakes on the car. Also, determine the initial momentum of the car before braking.(5 Marks)

Hint: Think about Laws of Motion in real world

2. Discuss the crucial role of safety features like seatbelts and airbags in modern vehicles in the context of Newton's laws of motion. Explain in detail how these features help to mitigate injuries during a sudden collision or impact.(5 Marks)

Hint: Think about Laws of Motion in real world

3. Analyze how Newton's Third Law of Motion precisely explains the propulsion of a rocket into space. What happens to the total momentum of the rocket system (comprising the rocket and the expelled exhaust gases) during its launch?(5 Marks)

Hint: Think about Laws of Motion in real world

4. Derive the mathematical expression for the law of conservation of momentum for two colliding bodies. Assume two objects with masses m1 and m2, moving with initial velocities u1 and u2 respectively, collide and move with final velocities v1 and v2.(5 Marks)

Hint: Think about Laws of Motion in real world